MAT21-1

Linear Algebra II

Determinants, Eigenvalues, and Diagonalization

Finding the directions in which a linear map merely stretches.

Albert School — 34 hours

Preface

In Linear Algebra I you learned to read a matrix as a **linear map**: a function $f(x) = Ax$ that respects sums and scalings. You learned to solve $Ax = b$ by Gaussian elimination, to compute kernels and images, to extract a basis and a dimension, and — at the very end — to write the **same** map in different bases using a change-of-basis matrix. That last idea is the door into this course.

A matrix is a description of a linear map **relative to a chosen basis**. Change the basis and the numbers change, but the map does not. So a natural question arises: among all the matrices that describe one given map, is there a **simplest** one? Is there a basis in which the map becomes transparent?

For a large and important class of maps the answer is yes. The simplest possible matrix is a **diagonal** one, which merely stretches each coordinate axis by its own factor. The special directions that get stretched rather than rotated are the **eigenvectors**, and the stretching factors are the **eigenvalues**. Finding them is the central project of this course.

To get there we first need a single number that detects whether a matrix is invertible and measures how it scales volume: the **determinant**. From the determinant we build the **characteristic polynomial**, whose roots are the eigenvalues. We then assemble eigenvectors into a change-of-basis matrix P and obtain the diagonalization $A = PDP^{-1}$. Along the way the **Cayley–Hamilton theorem** gives a surprising algebraic shortcut, and we close with three applications where these tools do real work: counting walks in graphs, solving recurrence relations, and predicting the long-run behaviour of Markov chains.

Throughout, every new idea is introduced first on a small concrete example, then stated in general. Watch for the **Common pitfall** boxes — they flag the errors students make most often.

Contents

1 Determinants

1.1 A number that decides invertibility

Recall from Linear Algebra I that a square matrix A is invertible exactly when the system $Ax = 0$ has only the trivial solution $x = 0$ — equivalently, when the columns of A are linearly independent. Checking this by row reduction works, but it gives a yes/no answer only after a full elimination. The **determinant** packs that decision into a single number $\det(A)$, with the rule:

$$A \text{ is invertible} \quad \Longleftrightarrow \quad \det(A) \neq 0.$$

Let us first see what this number is for small matrices, then understand **why** it detects invertibility.

For a 2×2 matrix the formula is

$$\det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

> **Worked example.** For $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ we get $\det(A) = 1 \cdot 4 - 2 \cdot 3 = 4 - 6 = -2 \neq 0$, so A is invertible. For $B = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ we get $\det(B) = 1 \cdot 4 - 2 \cdot 2 = 0$: the second column is twice the first, the columns are dependent, and B is **not** invertible.

1.2 Geometric meaning: signed volume

The determinant is not an arbitrary combination of entries. It measures the **signed area** (in 2D) or **signed volume** (in 3D and beyond) of the shape onto which the map $x \mapsto Ax$ sends the unit square (cube).

Take the two columns of a 2×2 matrix as vectors in the plane. They span a parallelogram. The absolute value $|\det(A)|$ is the area of that parallelogram, and the sign records orientation: positive if the columns keep the usual counter-clockwise orientation, negative if the map flips it.

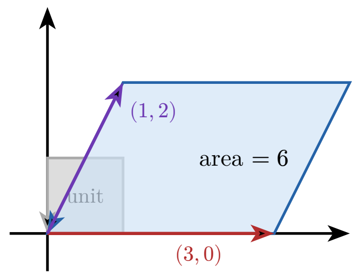

Figure 1: The map $A = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}$ sends the unit square (grey) to the parallelogram (blue) spanned by its columns $(3, 0)$ and $(1, 2)$. Its area is $|\det(A)| = |3 \cdot 2 - 1 \cdot 0| = 6$, so the map multiplies every area by 6.

This is the key intuition: **the determinant is the factor by which the map scales volume.** If $\det(A) = 0$ the parallelogram is flattened onto a line (zero area): the columns are dependent and the map is not invertible. If $\det(A) < 0$, the map also reverses orientation (like a mirror).

> **Remark** The "signed volume" picture also explains multiplicativity, below: applying two maps in succession scales volume by each factor in turn, so the volume factors multiply.

1.3 Computing 3×3 determinants: Sarrus' rule

For a 3×3 matrix there is a convenient mnemonic, **Sarrus' rule**: copy the first two columns to the right, then add the three "down-right" diagonal products and subtract the three "down-left" ones.

$$\det \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = aei + bfg + cdh - ceg - afh - bdi.$$

> **Worked example.**
>
> $$\det \begin{pmatrix} 2 & 0 & 1 \\ 1 & 3 & 2 \\ 0 & 1 & 1 \end{pmatrix} = 2 \cdot 3 \cdot 1 + 0 \cdot 2 \cdot 0 + 1 \cdot 1 \cdot 1 - 1 \cdot 3 \cdot 0 - 2 \cdot 2 \cdot 1 - 0 \cdot 1 \cdot 1$$
>
> $$= 6 + 0 + 1 - 0 - 4 - 0 = 3.$$

> **Common pitfall** Sarrus' rule works **only** for 3×3 matrices. There is no analogous "diagonals" trick for 4×4 or larger — the most common determinant error is to invent one. For larger matrices use cofactor expansion or row reduction.

1.4 Cofactor expansion (Laplace expansion)

The general definition works by **expanding along a row or column**. Fix a row i. For each entry a_{ij}, delete row i and column j to get the $(n-1) \times (n-1)$ **minor** M_{ij}, and define the **cofactor** $C_{ij} = (-1)^{i+j} \det(M_{ij})$. Then

$$\det(A) = \sum_{j=1}^{n} a_{ij} C_{ij} = \sum_{j=1}^{n} (-1)^{i+j} a_{ij} \det(M_{ij}).$$

The sign pattern $(-1)^{i+j}$ forms a checkerboard:

$$\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}.$$

> **Worked example.** Expand $\det \begin{pmatrix} 2 & 0 & 1 \\ 1 & 3 & 2 \\ 0 & 1 & 1 \end{pmatrix}$ along the **first row** (chosen because of the 0, which kills a term):
>
> $$= 2 \cdot (-1)^{1+1} \det \begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix} + 0 \cdot (\dots) + 1 \cdot (-1)^{1+3} \det \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$$
>
> $$= 2(3-2) + 1(1-0) = 2 + 1 = 3,$$
>
> matching the Sarrus computation above.

> **Choose the row or column with the most zeros.** Each zero entry removes an entire $(n-1) \times (n-1)$ determinant from the work. This is the single most effective hand-computation tactic.

A determinant can also be expanded down any column by the same formula with the roles of i and j swapped; the result is always the same number.

1.5 Computing by row reduction

Cofactor expansion costs roughly $n!$ multiplications — hopeless for large n. **Row reduction** is far cheaper, provided we track how each elementary operation changes the determinant:

Proposition 1 (Effect of row operations) Let A be square.
Swap two rows det changes sign.
Multiply a row by a scalar k det is multiplied by k.
Add a multiple of one row to another det is **unchanged**.

Reduce A to an upper-triangular matrix U using only these operations, keeping a running record of swaps and scalings. The determinant of a triangular matrix is just the product of its diagonal entries:

$$\det(\text{triangular}) = \prod_{i=1}^{n} u_{ii}.$$

Worked example. Compute $\det\begin{pmatrix} 2 & 4 & 2 \\ 1 & 3 & 2 \\ 1 & 1 & 3 \end{pmatrix}$.

Swap $R_1 \leftrightarrow R_2$ (sign flips, factor -1):

$$\begin{pmatrix} 1 & 3 & 2 \\ 2 & 4 & 2 \\ 1 & 1 & 3 \end{pmatrix}.$$

Now $R_2 \leftarrow R_2 - 2R_1$ and $R_3 \leftarrow R_3 - R_1$ (no change):

$$\begin{pmatrix} 1 & 3 & 2 \\ 0 & -2 & -2 \\ 0 & -2 & 1 \end{pmatrix}.$$

Then $R_3 \leftarrow R_3 - R_2$ (no change):

$$\begin{pmatrix} 1 & 3 & 2 \\ 0 & -2 & -2 \\ 0 & 0 & 3 \end{pmatrix}.$$

The triangular product is $1 \cdot (-2) \cdot 3 = -6$. Undo the one swap by multiplying by -1:

$$\det(A) = (-1) \cdot (-6) = 6.$$

Common pitfall When you scale a row to make a pivot equal to 1, you must **divide** the final product by that same scalar to compensate. Forgetting the bookkeeping for swaps (sign) and scalings (factor) is the classic row-reduction determinant mistake. The safest habit: use only "add a multiple of a row" operations plus the occasional swap, never scale.

1.6 Properties of the determinant

These properties follow from the definition (and the volume picture) and are used constantly.

Proposition 2 (Key properties) For square matrices A, B of the same size:

1. **Triangular / diagonal:** det is the product of the diagonal entries.
2. **Transpose:** $\det(A^T) = \det(A)$. (So every row statement has a column twin.)
3. **Multiplicativity:** $\det(AB) = \det(A)\det(B)$.
4. **Invertibility:** A is invertible $\iff \det(A) \neq 0$, and then $\det(A^{-1}) = \frac{1}{\det(A)}$.
5. **Scalar:** $\det(kA) = k^n \det(A)$ for an $n \times n$ matrix.
6. A matrix with a zero row, a repeated row, or two proportional rows has determinant 0.

Proof (of the inverse formula). If A is invertible then $AA^{-1} = I$. Taking determinants and using multiplicativity, $\det(A)\det(A^{-1}) = \det(I) = 1$, so $\det(A^{-1}) = \frac{1}{\det(A)}$. In particular $\det(A) \neq 0$. □

Common pitfall The determinant is **not additive**: in general $\det(A+B) \neq \det(A) + \det(B)$. For example with $A = B = I_2$, $\det(A+B) = \det(2I_2) = 4$ but $\det(A) + \det(B) = 1 + 1 = 2$. Only the **product** rule holds.

Exercises

1. Compute $\det\begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix}$ and $\det\begin{pmatrix} 4 & 8 \\ 1 & 2 \end{pmatrix}$; for each, say whether the matrix is invertible.

2. Use cofactor expansion along the column with the most zeros to compute

$$\det\begin{pmatrix} 1 & 0 & 2 \\ 3 & 0 & 4 \\ 5 & 6 & 7 \end{pmatrix}.$$

3. Compute $\det\begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{pmatrix}$ by row reduction, carefully tracking the effect of each operation.

4. Given that $\det(A) = 5$ for a 3×3 matrix, find $\det(2A)$, $\det(A^T)$, $\det(A^{-1})$, and $\det(A^3)$.

5. Show that if a square matrix has two identical rows, its determinant is 0. (Hint: what does swapping those two rows do?)

2 Eigenvalues and eigenvectors

2.1 The special directions of a map

Most vectors, when hit by a matrix A, change both length and direction. But some special nonzero vectors are merely **scaled**: their direction is preserved (or exactly reversed). These are the eigenvectors, and they reveal the skeleton of the transformation.

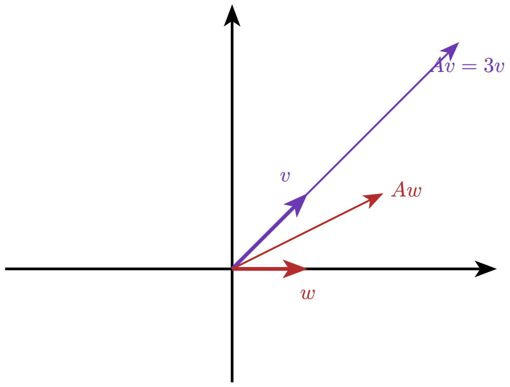

Figure 2: For $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$: the eigenvector $v = (1, 1)$ is stretched along its own line ($Av = 3v$), while the ordinary vector $w = (1, 0)$ is rotated off its line ($Aw = (2, 1)$ points in a new direction).

Definition 3 (Eigenvalue, eigenvector) Let A be an $n \times n$ matrix. A scalar λ is an **eigenvalue** of A if there exists a **nonzero** vector v with

$$Av = \lambda v.$$

Such a v is an **eigenvector** of A associated with λ.

Common pitfall The vector v must be **nonzero**. Since $A \cdot 0 = \lambda \cdot 0$ holds for every λ, allowing $v = 0$ would make every scalar an "eigenvalue" and the notion useless. By contrast, $\lambda = 0$ **is** an allowed eigenvalue — it occurs exactly when A has a nontrivial kernel, i.e. when A is not invertible.

2.2 The characteristic polynomial

How do we **find** the eigenvalues? Rewrite the defining equation:

$$Av = \lambda v \quad \Longleftrightarrow \quad Av - \lambda v = 0 \quad \Longleftrightarrow \quad (A - \lambda I)v = 0.$$

We need a **nonzero** v in the kernel of $A - \lambda I$. That happens exactly when $A - \lambda I$ is not invertible, i.e. when its determinant vanishes.

Definition 4 (Characteristic polynomial) The **characteristic polynomial** of an $n \times n$ matrix A is

$$\chi_A(\lambda) = \det(A - \lambda I).$$

It is a polynomial of degree n in λ, and its roots are exactly the eigenvalues of A.

Worked example. For $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$,

$$\chi_A(\lambda) = \det \begin{pmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{pmatrix} = (2 - \lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3).$$

The eigenvalues are $\lambda = 1$ and $\lambda = 3$ — exactly the stretch factors we saw geometrically in Figure 2.

2.3 Eigenspaces: finding the eigenvectors

Once an eigenvalue λ is known, its eigenvectors are the nonzero solutions of $(A - \lambda I)v = 0$. Together with 0 they form a subspace.

Definition 5 (Eigenspace) The **eigenspace** of A for the eigenvalue λ is

$$E_\lambda = \ker(A - \lambda I) = \{v : Av = \lambda v\}.$$

It is a subspace of $\mathbb{R}^n$; its nonzero vectors are the eigenvectors for λ. Its dimension is at least 1.

Worked example. Continue with $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.

For $\boldsymbol{\lambda = 3}$: solve $(A - 3I)v = 0$, i.e. $\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} v = 0$. The single equation is $-v_1 + v_2 = 0$, so $v_1 = v_2$ and $E_3 = \operatorname{span}\{(1, 1)\}$.

For $\boldsymbol{\lambda = 1}$: solve $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} v = 0$, i.e. $v_1 + v_2 = 0$, so $E_1 = \operatorname{span}\{(1, -1)\}$.

Both eigenspaces are lines (dimension 1).

Common pitfall Always substitute **one specific** eigenvalue into $A - \lambda I$ before solving. A frequent error is to keep λ symbolic and try to "solve" the system in general, or to forget that an eigenvector must be nonzero — the zero solution is never an eigenvector.

2.4 Trace and determinant from the eigenvalues

Two quick sanity checks come from comparing χ_A to its roots. Write the eigenvalues (with repetition) as $\lambda_1, \ldots, \lambda_n$. Then

$$\det(A) = \prod_{i=1}^{n} \lambda_i, \qquad \operatorname{tr}(A) = \sum_{i=1}^{n} \lambda_i,$$

where the **trace** $\operatorname{tr}(A)$ is the sum of the diagonal entries of A. So the determinant is the **product** of the eigenvalues and the trace is their **sum**.

Worked example. For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ the eigenvalues are 1 and 3. Indeed $\operatorname{tr}(A) = 2 + 2 = 4 = 1 + 3$ and $\det(A) = 3 = 1 \cdot 3$. These two equations are a cheap way to catch arithmetic slips before going further.

Remark For a 2×2 matrix this gives a memorable shortcut: $\chi_A(\lambda) = \lambda^2 - \operatorname{tr}(A)\lambda + \det(A)$.

Exercises

6. Find the characteristic polynomial, eigenvalues, and a basis of each eigenspace for $A = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}$.
7. Show that $\lambda = 0$ is an eigenvalue of A if and only if A is not invertible.
8. A matrix has eigenvalues $2, 5, -1$. What are its trace and determinant?

9. Find the eigenvalues of the triangular matrix $\begin{pmatrix} 3 & 7 & 1 \\ 0 & 3 & 4 \\ 0 & 0 & -2 \end{pmatrix}$. (What do you notice about the diagonal of a triangular matrix?)
10. Verify that if $Av = \lambda v$, then $A^2 v = \lambda^2 v$. What is an eigenvalue of A^2 in terms of those of A?

3 Multiplicities and diagonalization

3.1 Algebraic vs. geometric multiplicity

An eigenvalue can be a **repeated** root of χ_A, and the eigenspace it produces can have various sizes. We need two separate counts.

Definition 6 (The two multiplicities) Let λ be an eigenvalue of A.

Algebraic multiplicity $m_a(\lambda)$ the number of times $(\lambda - \lambda_0)$ divides χ_A — i.e. how often it appears as a root.

Geometric multiplicity $m_g(\lambda)$ the dimension of the eigenspace E_λ — i.e. the number of independent eigenvectors for λ.

These always satisfy

$$1 \le m_g(\lambda) \le m_a(\lambda).$$

The gap between them is precisely what can prevent diagonalization.

Worked example — When they differ. Consider the shear $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$. Its characteristic polynomial is $\chi_A(\lambda) = (2-\lambda)^2$, so $\lambda = 2$ has algebraic multiplicity 2. But

$$A - 2I = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},$$

whose kernel is $\operatorname{span}\{(1,0)\}$ — only **one** dimension. So $m_g(2) = 1 < 2 = m_a(2)$. There is a "missing" eigenvector. This matrix will turn out to be **not diagonalizable**.

Common pitfall A repeated eigenvalue does **not** automatically have too few eigenvectors. The matrix $2I = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$ also has $\chi(\lambda) = (2-\lambda)^2$, but its eigenspace is all of $\mathbb{R}^2$ ($m_g = 2$). You must actually compute $\dim\ker(A - \lambda I)$ — never assume it from the algebraic multiplicity.

3.2 Why diagonalization, and what it is

Suppose we can find a **basis of $\mathbb{R}^n$ made entirely of eigenvectors** $v_1, \dots, v_n$, with $Av_i = \lambda_i v_i$. Place these eigenvectors as the columns of a matrix P, and put the eigenvalues on the diagonal of D:

$$P = \begin{pmatrix} | & & | \\ v_1 & \cdots & v_n \\ | & & | \end{pmatrix}, \quad D = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}.$$

Then $AP = PD$ (each column says $Av_i = \lambda_i v_i$), and since the columns of P are a basis, P is invertible. Therefore

$$A = PDP^{-1}.$$

This is **diagonalization**. Reading it right to left, P^{-1} rewrites a vector in the eigenvector basis, D scales each eigen-coordinate, and P translates back. In the eigenvector basis the map is just independent stretches.

Definition 7 A square matrix A is **diagonalizable** if $A = PDP^{-1}$ for some invertible P and diagonal D — equivalently, if $\mathbb{R}^n$ has a basis of eigenvectors of A.

3.3 The diagonalizability criterion

Theorem 8 (Diagonalizability criterion) An $n \times n$ matrix A is diagonalizable (over $\mathbb{R}$) if and only if both:

1. the characteristic polynomial **splits** into linear factors over $\mathbb{R}$ (all n roots are real, counted with multiplicity); and
2. for **every** eigenvalue, $m_g(\lambda) = m_a(\lambda)$.

Equivalently, the geometric multiplicities sum to n, so there are enough independent eigenvectors to form a basis.

A useful sufficient (not necessary) shortcut:

Corollary 9 If an $n \times n$ matrix has n **distinct** real eigenvalues, it is diagonalizable. (Each then has $m_a = m_g = 1$ automatically.)

Worked example — A full diagonalization. Diagonalize $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$. From earlier, eigenvalues 3 and 1 are distinct, with eigenvectors $(1, 1)$ and $(1, -1)$. Take

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad D = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}.$$

Then $P^{-1} = -\frac{1}{2}\begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, and one checks $A = PDP^{-1}$.

Worked example — A matrix that fails. The shear $\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$ has $m_g(2) = 1 < 2 = m_a(2)$, so condition 2 fails: it is **not diagonalizable**. There simply is no basis of $\mathbb{R}^2$ made of its eigenvectors.

Common pitfall Distinct eigenvalues **guarantee** diagonalizability, but their absence proves nothing. With a repeated eigenvalue you must check the geometric multiplicity before concluding either way.

3.4 Matrix powers via diagonalization

The payoff. Once $A = PDP^{-1}$, powers collapse beautifully because the inner $P^{-1}P$ pairs cancel:

$$A^2 = PDP^{-1}PDP^{-1} = PD^2P^{-1}, \quad \text{and in general} \quad A^n = PD^nP^{-1}.$$

Powering a diagonal matrix is trivial — just raise each diagonal entry: $D^n = \mathrm{diag}(\lambda_1^n, \ldots, \lambda_n^n)$.

Worked example. With $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = PDP^{-1}$ as above,

$$A^n = P\begin{pmatrix} 3^n & 0 \\ 0 & 1 \end{pmatrix}P^{-1} = \frac{1}{2}\begin{pmatrix} 3^n+1 & 3^n-1 \\ 3^n-1 & 3^n+1 \end{pmatrix}.$$

No repeated matrix multiplication — a closed form for every power at once.

3.5 Application: linear recurrence relations

Diagonalization turns a recurrence into a power, and the power into a formula. The classic example is the **Fibonacci sequence** $F_0 = 0$, $F_1 = 1$, $F_{n+1} = F_n + F_{n-1}$. Stack two consecutive terms into a vector and notice the update is linear:

$$\begin{pmatrix} F_{n+1} \\ F_n \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} F_n \\ F_{n-1} \end{pmatrix} = A\begin{pmatrix} F_n \\ F_{n-1} \end{pmatrix}, \quad \text{so} \quad \begin{pmatrix} F_{n+1} \\ F_n \end{pmatrix} = A^n\begin{pmatrix} F_1 \\ F_0 \end{pmatrix}.$$

Diagonalizing $A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ (eigenvalues $\varphi = \frac{1+\sqrt{5}}{2}$ and $\psi = \frac{1-\sqrt{5}}{2}$) and computing A^n yields **Binet's formula**

$$F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}.$$

The eigenvalues **are** the growth rates of the sequence; the dominant one, φ, governs the long-run behaviour.

Exercises

11. For $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$: find the eigenvalues and eigenvectors, write a diagonalization $A = PDP^{-1}$, and use it to compute A^n.

12. For each matrix decide whether it is diagonalizable, justifying via the two multiplicities: $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix}$, $\begin{pmatrix} 3 & 0 \\ 1 & 3 \end{pmatrix}$.

13. The matrix $A = \begin{pmatrix} 4 & 0 & 0 \\ 1 & 4 & 0 \\ 0 & 0 & 2 \end{pmatrix}$ has eigenvalues $4, 4, 2$. Compute the geometric multiplicity of $\lambda = 4$ and decide whether A is diagonalizable.

14. Solve the recurrence $a_{n+1} = 3a_n - 2a_{n-1}$ with $a_0 = 1$, $a_1 = 3$ by diagonalizing $\begin{pmatrix} 3 & -2 \\ 1 & 0 \end{pmatrix}$ and finding a closed form for a_n.

4 Cayley–Hamilton and the matrix inverse

4.1 A matrix satisfies its own characteristic polynomial

There is a striking algebraic fact linking a matrix to its characteristic polynomial.

Theorem 10 (Cayley–Hamilton) Every square matrix A satisfies its own characteristic equation: if $\chi_A(\lambda) = \lambda^n + c_{n-1}\lambda^{n-1} + \dots + c_1\lambda + c_0$, then substituting A for λ (and $c_0 I$ for the constant) gives the zero matrix:

$$\chi_A(A) = A^n + c_{n-1}A^{n-1} + \dots + c_1 A + c_0 I = 0.$$

Common pitfall Substitute the **matrix** A, not a number, and replace the constant term c_0 by $c_0 I$ — you cannot add a scalar to a matrix. Writing "$\chi_A(A) = ... + c_0$" (a scalar) instead of "$... + c_0 I$" is the standard slip. Also note $\chi_A(\lambda) = \det(A - \lambda I)$ is a **polynomial**; the theorem is **not** the trivial statement $\det(A - AI) = 0$.

Worked example — Verification. For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ we found $\chi_A(\lambda) = \lambda^2 - 4\lambda + 3$. Check $A^2 - 4A + 3I$:

$$A^2 = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}, \quad 4A = \begin{pmatrix} 8 & 4 \\ 4 & 8 \end{pmatrix}, \quad 3I = \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix},$$

$$A^2 - 4A + 3I = \begin{pmatrix} 5-8+3 & 4-4+0 \\ 4-4+0 & 5-8+3 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}. \checkmark$$

4.2 The inverse as a polynomial in A

Cayley–Hamilton gives a slick way to express A^{-1} — **when** A is invertible — using only powers of A, no Gaussian elimination. Start from $\chi_A(A) = 0$ and isolate the constant term. For the 2×2 case, $A^2 - \operatorname{tr}(A)A + \det(A)I = 0$, so

$$A^2 - \operatorname{tr}(A)A = -\det(A)I \quad \Longrightarrow \quad A(A - \operatorname{tr}(A)I) = -\det(A)I.$$

If $\det(A) \neq 0$, divide:

$$A^{-1} = \frac{1}{\det(A)}(\operatorname{tr}(A)I - A).$$

Worked example. For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$: $\operatorname{tr}(A) = 4$, $\det(A) = 3$, so

$$A^{-1} = \frac{1}{3}(4I - A) = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

Indeed $AA^{-1} = I$.

This generalizes: from $\chi_A(A) = 0$ one always solves for $c_0 I = -A(...)$ and, since $c_0 = (-1)^n \det(A) \neq 0$ for invertible A, divides through to write A^{-1} as a polynomial in A.

4.3 Powers of non-diagonalizable matrices

Cayley–Hamilton also tames matrices that **cannot** be diagonalized, where the $A^n = PD^nP^{-1}$ trick is unavailable. The relation $\chi_A(A) = 0$ lets you rewrite A^n for any large n in terms of lower powers $I, A, ..., A^{n-1}$, reducing the degree.

Worked example. The shear $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$ is not diagonalizable, but $\chi_A(\lambda) = (\lambda - 2)^2 = \lambda^2 - 4\lambda + 4$, so $A^2 = 4A - 4I$. Every higher power reduces to the form $\alpha A + \beta I$; one finds the pattern

$$A^n = \begin{pmatrix} 2^n & n2^{n-1} \\ 0 & 2^n \end{pmatrix}.$$

4.4 Computing the inverse by Gaussian elimination

For a general $n \times n$ matrix the most reliable hand method for the inverse is **Gauss–Jordan elimination on the augmented matrix** $[A \mid I]$. Row-reduce until the left block becomes I; the right block is then A^{-1}:

$$[A \mid I] \xrightarrow{\text{row reduce}} [I \mid A^{-1}].$$

The reasoning: each row operation is left-multiplication by an elementary matrix E; the sequence multiplies the left block to $E_k \cdots E_1 A = I$, so $E_k \cdots E_1 = A^{-1}$, which is exactly what the same operations build on the right block starting from I.

Worked example. Invert $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.

$$[A \mid I] = \left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 4 & 0 & 1 \end{array}\right).$$

$R_2 \leftarrow R_2 - 3R_1$:

$$\left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & -2 & -3 & 1 \end{array}\right).$$

$R_2 \leftarrow -\frac{1}{2}R_2$:

$$\left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{array}\right).$$

$R_1 \leftarrow R_1 - 2R_2$:

$$\left(\begin{array}{cc|cc} 1 & 0 & -2 & 1 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{array}\right).$$

Hence $A^{-1} = \begin{pmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{pmatrix}$.

Common pitfall If the left block ever produces a zero row before reaching I, then A is **not invertible** (its determinant is 0) and the process simply stops — there is no inverse to find.

Exercises

15. State χ_A for $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and verify Cayley–Hamilton by direct computation.

16. Use Cayley–Hamilton to express A^{-1} as a polynomial in A for $A = \begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix}$, then check by computing AA^{-1}.

17. Invert $\begin{pmatrix} 2 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$ by Gauss–Jordan elimination on $[A \mid I]$.

18. For the non-diagonalizable $A = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}$, use $\chi_A(A) = 0$ to express A^3 in the form $\alpha A + \beta I$.

5 Applications: graphs and Markov chains

5.1 Similar matrices: same map, different basis

Diagonalization is one instance of a broader relationship. Two matrices A and B are **similar** if $B = P^{-1}AP$ for some invertible P. From Linear Algebra I you know that changing the basis of a linear map replaces its matrix exactly this way — so **similar matrices represent the same linear map in different coordinate systems.**

Proposition 11 (Similarity invariants) Similar matrices share the same characteristic polynomial, and hence the same eigenvalues, determinant, and trace.

This is **why** diagonalization is meaningful: $A = PDP^{-1}$ says the map looks like the simple diagonal map D once we choose the eigenvector basis encoded by P. The eigenvalues are intrinsic to the map; the messy entries of A are an artefact of the standard basis. Diagonalizing is choosing the coordinate system in which the map is as simple as it can be.

5.2 Counting walks with adjacency matrices

Let a graph on vertices $1, \ldots, n$ have **adjacency matrix** A, where $A_{ij} = 1$ if there is an edge from i to j and 0 otherwise. A beautiful fact connects matrix powers to the graph's structure.

Theorem 12 (Walk counting) The (i, j) entry of A^k equals the number of **walks of length k** (sequences of k edges) from vertex i to vertex j.

Proof. By induction. For $k = 1$, A_{ij} counts the single-edge walks (it is 1 or 0). For the step, a walk of length $k+1$ from i to j is a walk of length k from i to some intermediate vertex m followed by an edge $m \to j$. Summing over m, $\left(A^{k+1}\right)_{ij} = \sum_m \left(A^k\right)_{im} A_{mj}$ — which is exactly the definition of matrix multiplication $\left(A^k A\right)_{ij}$. $\square$

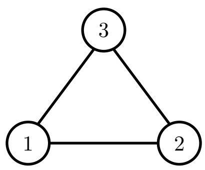

Figure 3: A triangle graph on vertices $1, 2, 3$ with adjacency matrix $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$. The number of walks of length k between any two vertices is read off from A^k.

Worked example. For the triangle of Figure 3,

$$A^2 = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$$

Entry $(1, 1) = 2$: there are two walks of length 2 from vertex 1 back to itself ($1 \to 2 \to 1$ and $1 \to 3 \to 1$). Entry $(1, 2) = 1$: one walk of length 2 from 1 to 2 (namely $1 \to 3 \to 2$).

The **dominant eigenvalue** of A (the largest in absolute value) controls how the walk counts grow as $k \to \infty$, and its eigenvector gives the asymptotic **direction** — the relative likelihood of ending at each vertex. This is the bridge to our final topic.

5.3 Markov chains and stochastic matrices

A **Markov chain** models a system that hops between states with fixed probabilities. Collect the transition probabilities into a **column-stochastic matrix** P, where P_{ij} is the probability of moving **to** state i **from** state j. Each column sums to 1 (from any state, the probabilities of where you go next add to one).

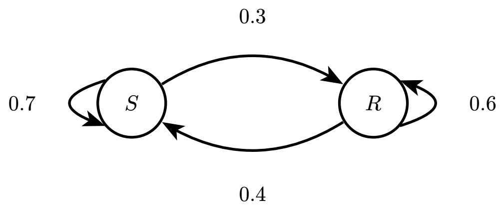

Figure 4: A two-state weather chain: Sunny (S) and Rainy (R). From sunny, stay sunny with probability 0.7 or turn rainy with 0.3; from rainy, stay rainy with 0.6 or turn sunny with 0.4.

For Figure 4, ordering the states (S, R), the column-stochastic matrix is

$$P = \begin{pmatrix} 0.7 & 0.4 \\ 0.3 & 0.6 \end{pmatrix}.$$

A probability distribution $x_k = \binom{p_S}{p_R}$ (entries ≥ 0 summing to 1) evolves by $x_{k+1} = Px_k$, so after k steps $x_k = P^k x_0$.

> **Definition 13 (Stationary distribution)** A **stationary distribution** is a probability vector π with $P\pi = \pi$ — i.e. an **eigenvector of P for the eigenvalue 1**, normalized so its entries sum to 1. It is the distribution that no longer changes from step to step.

Every column-stochastic matrix has 1 as an eigenvalue (because the all-ones row vector is a left eigenvector: each column sums to 1). For our weather chain, solve $(P - I)\pi = 0$:

$$\begin{pmatrix} -0.3 & 0.4 \\ 0.3 & -0.4 \end{pmatrix}\pi = 0 \quad\Longrightarrow\quad 3p_S = 4p_R.$$

With $p_S + p_R = 1$ this gives $\pi = \binom{\frac{4}{7}}{\frac{3}{7}} \approx \binom{0.571}{0.429}$: in the long run it is sunny about 57% of the time.

5.4 Convergence via diagonalization

Why does the chain **settle** to π regardless of where it starts? Diagonalize. The eigenvalues of $P = \begin{pmatrix} 0.7 & 0.4 \\ 0.3 & 0.6 \end{pmatrix}$ are $\lambda_1 = 1$ and $\lambda_2 = 0.3$ (their product is $\det P = 0.42 - 0.12 = 0.3$ and their sum is $\operatorname{tr} P = 1.3$). Write any starting distribution in the eigenvector basis:

$$x_0 = c_1 v_1 + c_2 v_2, \quad \text{with } v_1 = \pi.$$

Then

$$x_k = P^k x_0 = c_1 \lambda_1^k v_1 + c_2 \lambda_2^k v_2 = c_1 \cdot 1^k \cdot \pi + c_2 \cdot (0.3)^k \cdot v_2.$$

Because $|\lambda_2| = 0.3 < 1$, the second term decays to 0 as $k \to \infty$, and $x_k \to c_1\pi$. (One checks $c_1 = 1$ for any probability start.) So

$$\lim_{k\to\infty} x_k = \pi \quad \text{for every initial distribution } x_0.$$

This is the whole story of long-run behaviour as a **linear-algebra** fact: the eigenvalue 1 supplies the surviving stationary direction, and every other eigenvalue has $|\lambda| < 1$, so its contribution dies away. The dominant eigenvalue dictates the limit; the rest set the **speed** of convergence.

Common pitfall Be consistent about convention. With **column**-stochastic P as here, columns sum to 1 and distributions are columns updated by $x_{k+1} = Px_k$. Many texts use **row**-stochastic matrices (rows sum to 1) with row-vector distributions updated by $x_{k+1} = x_k P$. The two are transposes of each other — fix one convention and keep your dimensions consistent.

Exercises

19. For the graph with adjacency matrix $A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$ (a path $1 - 2 - 3$), compute A^2 and A^3 and interpret the entries as walk counts.

20. Show that if A is the adjacency matrix of a graph, then $\left(A^k\right)_{ii}$ counts the closed walks of length k starting and ending at vertex i.

21. A frog hops between two lily pads. From pad 1 it stays with probability 0.5 and jumps to pad 2 with 0.5; from pad 2 it stays with 0.8 and jumps to pad 1 with 0.2. Write the column-stochastic matrix and find the stationary distribution.

22. For the matrix P in the previous exercise, find both eigenvalues and explain, using diagonalization, why the distribution converges to the stationary one from any start. Roughly how fast does it converge?

23. Explain why two similar matrices must have the same eigenvalues, by comparing their characteristic polynomials.